Physics-Informed Graph Neural Networks with Uncertainty Quantification for Dynamic Valuation of Blue Carbon Assets

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***Abstract*—Voluntary carbon markets price mangrove non-permanence risk with flat, rule-based buffer pools (15–20% under Verra VM0033 and Plan Vivo) applied uniformly regardless of site-level uncertainty. We test whether a physics-informed graph neural network (GNN) combined with conformalized quantile regression (CQR) can support a differentiated alternative, on one pilot region: the Ten Thousand Islands mangrove fringe, southwest Florida. We build a 10,124-node spatiotemporal graph from Sentinel-2 imagery, Global Mangrove Watch extent, and an independently field- calibrated soil organic carbon (SOC) product, and train a GraphSAGE regressor with an added mass-balance-inspired loss (an NDVI accretion proxy and a tide-gauge-distance export proxy) against a plain GNN and a linear regression baseline, on an identical spatially-blocked train/validation/test split. The physics-informed model improves test R^2 over the no-physics GNN (-0.383 to -0.219), but both underperform linear regression ($+0.031$); its predictions correlate with the imposed physical prior at 0.645 on held-out nodes, versus 0.048 and -0.060 for the non-physics models, evidence the constraint transfers under spatial shift, not that it has been independently validated against physical measurements. CQR, calibrated on a held-out validation block, achieves 85.7% and 90.2% empirical coverage at 90% and 95% nominal levels, undershooting both. A proof-of-concept pricing mechanism maps interval width to a per-node multiplier ranging 0.50–1.50 across five representative test nodes, against one fixed 0.825 multiplier under a flat 17.5% buffer. Point-prediction accuracy remains weak under spatial distribution shift in this pilot; the paper's contribution is a calibrated, honestly-reported uncertainty signal, not point accuracy, as the basis for differentiated pricing.**

***Index Terms*—blue carbon, graph neural network, physics-informed learning, conformal prediction, uncertainty quantification, carbon credit pricing**

I. Introduction

Voluntary carbon credit markets discount mangrove restoration and conservation credits for non-permanence risk (the possibility that stored carbon is later released, through erosion, sea-level rise, or disturbance) using a buffer pool: a fixed percentage of issued credits withheld from sale. Verra's VM0033 methodology and the Plan Vivo Standard both apply a flat 15–20% buffer, set by a checklist-based risk score and revisited only periodically (Plan Vivo: roughly every ten years), applied uniformly across an entire project regardless of how uncertainty varies within it [1], [2]. A restored interior mangrove stand and an eroding tidal fringe within the same project receive the same discount.

This paper asks whether a machine-learned, per-site uncertainty signal can support a differentiated alternative. We combine two components. First, a physics-informed graph neural network (GNN): mangrove patches are represented as nodes in a spatial graph, and a soft loss term encodes a simplified carbon mass-balance prior, following the general approach of physics-informed neural networks [3] and its graph-based extensions [4], which have been applied to mass-conserving flood forecasting [5], [6] but not, to our knowledge, to blue carbon. Second, conformalized quantile regression (CQR) [7], which gives finite-sample coverage guarantees under an exchangeability assumption we discuss and test directly, rather than the heuristic uncertainty of Monte Carlo dropout [8].

The scope is deliberately narrow: one pilot region (the Ten Thousand Islands mangrove fringe, southwest Florida), one physics constraint (a steady-state mass-balance proxy, not a full temporal flux equation), and one primary UQ method (CQR, with deep ensembles and MC dropout left as future ablations). We state the result plainly rather than building up to it: on this pilot's spatially-blocked evaluation, the physics-informed GNN does not out-predict a plain linear regression baseline. What it does do is generalize the imposed physical prior to spatially held-out nodes far more strongly than either non-physics model, and pairing that with a calibrated (if imperfectly so) uncertainty interval gives a differentiated pricing signal a flat buffer cannot produce. The paper's contribution is this evaluation and its honest limits, not a claim of validated point- prediction accuracy.

II. Related Work

A. Physics-informed graph neural networks

Physics-informed neural networks add a PDE-residual term to the training loss [3]. Graph Network Simulators encode physics architecturally, through local, permutation-invariant message passing, rather than through an explicit residual [4]. The closest prior art we found combining both – a spatiotemporal graph with an explicit conservation-law loss term – is in flood forecasting, where mass conservation across a river or floodplain graph is added directly to the training objective [5], [6]. Every such example we found operates on hydrological networks. None target mangrove biomass/soil-carbon dynamics, and none couple the physics-informed pre-

diction to a priced financial instrument through uncertainty quantification. That combination is this paper's delta.

B. Mangrove biomass and carbon stock estimation

The dominant approach in the remote-sensing literature is per-pixel tabular regression (random forest, XGBoost) on spectral indices, e.g. on Sentinel-2 and ALOS-2 imagery in Malaysia ($R^2 = 0.735$) and comparable studies elsewhere. Simard et al. produced a 30 m global mangrove aboveground biomass and canopy height product [9], the standard reference we could not access in this sandbox (it requires a NASA Earthdata login). Radabaugh et al. built a field-plot-calibrated, 30 m spatial model of above- and belowground carbon stocks for the southwest Florida coast, broken out by four named sub-regions, one of which – Ten Thousand Islands – matches our pilot AOI directly [10]. Sanderman et al. produced a global, field-calibrated 30 m soil organic carbon (SOC) product from over 10,000 field soil samples [11]; this is the label our models are trained against. None of this literature uses a graph representation or a physical constraint; that is the gap this paper's method addresses.

C. Buffer-pool non-permanence risk pricing

Verra's VM0033 methodology and the Plan Vivo Standard both discount credits using a flat, checklist-derived buffer percentage [1], [2], reassessed infrequently and applied uniformly within a project. This is the business-side baseline our pricing proof-of-concept (Section V) compares against.

D. Uncertainty quantification

Monte Carlo dropout [8] is cheap but known to be poorly calibrated, particularly under distribution shift. Deep ensembles [12] are better calibrated empirically but require training multiple full models. Conformalized quantile regression [7] gives a finite-sample, distribution-free coverage guarantee under an exchangeability assumption, and has been validated for earth-observation regression tasks [13]. We adopt CQR as the primary method (full comparison in our project's `uq_decision.md`), with the caveat, tested directly in Section IV, that spatial autocorrelation can violate the exchangeability assumption conformal prediction relies on.

III. Method

A. Pilot region and data

The pilot AOI is the Ten Thousand Islands / Rookery Bay mangrove fringe, southwest Florida (bounding box $-81.90°$ to $-81.40°$ longitude, $25.90°$ to $26.30°$ latitude), chosen because every required data layer is reachable without a paid account: Sentinel-2 L2A imagery via Microsoft Planetary Computer, Global Mangrove Watch (GMW) v3 extent [14], the Sanderman et al. SOC product [11], and a NOAA CO-OPS tide gauge (station 8725114, Naples Bay North) inside the AOI. Field-calibrated validation numbers come from Radabaugh et al. [10].

B. Graph construction

Native 30 m SOC grid cells are aggregated into 5×5 blocks ($\sim$150 m nodes); Sentinel-2 bands are resampled onto this same grid (bilinear) and the GMW mask (nearest-neighbor), so the regression label itself is never resampled. A node is kept if its block-mean GMW mangrove fraction exceeds 0.5 and it has a valid SOC value, giving 10,124 nodes and 64,366 directed edges (8-connectivity). Node features are NDVI, raw B04/B08/B11 reflectance, and Euclidean distance to the tide gauge; the label is mean SOC (t C/ha, 0–100 cm).

An initial spatial train/test split (five contiguous longitude bands, the middle band held out) produced negative test R^2 for every model family tried, including a naive mean predictor, because the held-out band happened to straddle a real regional carbon-density gradient – Radabaugh et al.'s own sub-regional figures range from 265.1 ± 43.2 to $409 \pm 104.4\,\mathrm{Mg\,C\,ha^{-1}}$ [10]. We replaced this with an 8×8 checkerboard grid of superblocks ($\sim$6 km each, far larger than the 150 m node spacing), assigning superblocks to train/validation/test by cumulative node count rather than superblock count (mangrove nodes are concentrated in a thin coastal fringe, so a uniform grid has very uneven occupancy). The final split is 5,464 train / 2,424 validation / 2,236 test nodes, with train and test label means close (404.6 and 403.3 t/ha respectively). Validation is used only for early stopping and CQR calibration, never for the reported test metric.

C. Architecture

All models share a 2-layer GraphSAGE backbone (hidden dimension 16, dropout 0.3), Adam optimizer (lr $= 0.01$, weight decay 10^{-3}), and early stopping on validation MSE (patience 30 epochs), seed 42 throughout. The baseline and physics-informed variants differ only in loss function, so any difference between them isolates the effect of the physics term.

D. Physics-informed loss

The full mass-balance relationship is $dC/dt = \text{Accretion} - \text{Export} - \text{Decomposition}$. A single Sentinel-2 date and no flux-tower data rule out estimating this directly. We impose a simplified steady-state prior instead: predicted stock is softly encouraged toward an equal-weighted, min-max-normalized (on training data only) combination of an accretion proxy (NDVI) and an export proxy (distance to the tide gauge, a positional stand-in for tidal energy in the absence of a per-node hydrodynamic layer). A second, unconditional term penalizes negative carbon predictions, a direct corollary of mass balance rather than an independent constraint. Both terms are combined into one physics loss with two fixed weights ($w_{\text{prior}} = 0.3$, $w_{\text{nonneg}} = 0.1$) and no fitted coefficients, added to the standard MSE data loss.

E. Uncertainty quantification: CQR

A quantile-head GraphSAGE (same backbone) is trained with pinball loss to predict the 2.5th, 5th, 95th, and 97.5th percentiles. Split-conformal calibration is performed on the validation set: for each nominal level $1 - \alpha$, the conformity

score $E_i = \max(q_{\text{lo}}(x_i) - y_i,\, y_i - q_{\text{hi}}(x_i))$ is computed on validation, and the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of $\{E_i\}$ adjusts the interval bounds. Empirical coverage is then measured once on the held-out test set. The quantile regressor is trained with the data loss only, not the physics loss; combining the two is left as future work.

IV. Experiments and Results

TABLE I
THREE-WAY MODEL COMPARISON (IDENTICAL SPLIT, SEED 42)

Model	Train R^2	Test R^2	Test RMSE (t/ha)
Linear regression	0.040	**0.031**	96.0
Baseline GNN (no physics)	0.246	-0.383	114.7
Physics-informed GNN	0.160	-0.219	107.7

Table I reports the central comparison. The physics term improves test R^2 over the no-physics GNN by $+0.165$ (-0.383 to -0.219) and validation R^2 by $+0.064$ (0.137 to 0.201), but neither GNN variant beats linear regression on this held-out spatial block. This result is reported as measured, not adjusted for.

TABLE II
PLAUSIBILITY METRICS ON THE TEST SPLIT

Model	Neg. rate	Edge incoh.	Prior-gen. corr.
Linear regression	0.0	0.00031	-0.060
Baseline GNN	0.0	0.0	$+0.048$
Physics-informed GNN	0.0	0.0	$\mathbf{+0.645}$

Table II's negative-prediction rate and edge- incoherence rate (fraction of neighboring-node prediction jumps exceeding the 95th percentile of real training-side neighbor differences) do not differentiate the physics-informed model: no model predicts negative carbon on this AOI, and both GNN variants score 0.0 on edge incoherence, a property of graph message-passing smoothing generically, not of the physics term specifically. The last column, prior-generalization correlation, does differentiate sharply: the physics-informed model's test-node predictions correlate with the mass-balance proxy at 0.645, against 0.048 and -0.060 for the non-physics models. **This number requires precise interpretation.** The proxy being correlated against is the exact quantity the physics loss trained the model toward, so this is not independent physical validation. It shows that the physics-informed model's predictions generalize the imposed mass-balance-inspired prior to spatially held-out nodes, at a materially higher correlation than either non-physics baseline – evidence the constraint is learned and transferred, not that it has been independently validated against ground-truth physical measurements, which this AOI's single-date, flux-tower-free data cannot provide. That the constraint transfers at all is not trivial: every other metric in Table I shows this AOI generalizing poorly under spatial holdout, so a signal that does carry over is informative.

Table III shows both nominal levels undershooting empirical coverage by 4.3–4.8 percentage points. Interval widths (262–310 t/ha) are 2.5–3$\times$ the label's own standard deviation (103.4 t/ha), so the undercoverage is on top of already-wide, not falsely narrow, intervals. The negative correction Q indicates the base quantile regressor was already conservative on the calibration set; test undercoverage despite this tightening indicates the validation and test splits' residual distributions genuinely differ, consistent with the spatial-exchangeability risk flagged for this method before any model was trained.

TABLE III
CQR COVERAGE, CALIBRATED ON VALIDATION, EVALUATED ONCE ON TEST

Nominal	Empirical coverage	Mean width (t/ha)	Correction Q
90%	85.7%	261.9	-3.99
95%	90.2%	310.0	-5.31

V. Pricing Proof-of-Concept

We illustrate, without validating, how a per-node uncertainty signal could differentiate pricing where a flat buffer cannot. For each test node, a relative interval width is computed as the 90%-CQR interval width divided by the test set's own median width (257.6 t/ha; range 114.8–425.4 t/ha). A price multiplier is $1 - \alpha(\text{relative width} - 1)$, clipped to $[0.5, 1.5]$, with $\alpha = 1.0$ an arbitrary illustrative sensitivity, not fit to any market outcome. Table IV shows five nodes spanning the test set's interval-width distribution (minimum, 25th, 50th, 75th percentile, maximum), against a flat 17.5% buffer (the midpoint of Verra/Plan Vivo's cited 15–20% range) applied uniformly.

TABLE IV
PRICING PROOF-OF-CONCEPT: 5 REPRESENTATIVE TEST NODES

Node	Interval width	Our multiplier	Flat-buffer mult.
Min width	114.8	1.500	0.825
p25	239.1	1.072	0.825
Median	257.6	1.000	0.825
p75	283.6	0.899	0.825
Max width	425.4	0.500	0.825

Across these five real test nodes, our multiplier spans a 100-percentage- point range (0.50 to 1.50) against one fixed value (0.825) under a flat buffer. This demonstrates the mechanism – that a real, per-node uncertainty signal can produce differentiated pricing – not a validated price. α, the clipping bounds, and the flat-buffer reference value are all stated, arbitrary choices, not calibrated against any market or outcome data.

VI. Limitations

Single-date imagery. Only one Sentinel-2 date was available this sprint (a scoping decision, not an access failure), so the physics term constrains a cross-sectional steady-state relationship, not an actual dC/dt. **No flux or decomposition data.** No sediment-trap, flux-tower, or decomposition-rate measurements exist for this AOI; the accretion and export terms are both proxies, not measurements. **Distance-to-gauge**

is a positional proxy. It is a single-point Euclidean distance, not a per-node tidal-energy or inundation-frequency layer, and functions partly as a stand-in for spatial position rather than a validated hydrological signal. **The prior-generalization result is not independent validation**, as stated in Section IV: correlating predictions against the physics loss's own training target measures transfer to held-out nodes, not physical correctness. **CQR was trained without the physics loss**, a disclosed scope choice, not a combined physics-UQ model. **CQR undercoverage** (85.7%/90.2% vs. 90%/95% nominal) indicates the spatial-exchangeability assumption is stressed in this AOI, consistent with the risk flagged for this method before any model was run; a spatially-aware conformal variant is future work, not implemented here. **Single seed, single region.** All results use seed 42 and one pilot AOI; no multi-seed robustness check or second region was run this sprint. **The pricing mechanism is illustrative**, with an unfit sensitivity parameter and no market validation.

VII. Conclusion

On one pilot region, a physics-informed loss measurably improves a GNN's test-set R^2 over a no-physics variant, but neither closes the gap to a plain linear baseline; both facts are reported as measured. The physics term's clearest effect is that its predictions carry the imposed mass-balance-inspired prior forward to spatially held-out nodes far more than either non-physics model does, evidence of transfer, not of physical validation, given this AOI's data cannot supply the latter. Conformal calibration gives usable, if imperfectly covered, intervals, and a simple mapping from interval width to price multiplier demonstrates that a per-node uncertainty signal can differentiate pricing where today's flat buffer pools cannot. Future work includes a second Sentinel-2 date to recover an actual temporal accretion signal, a spatially-aware conformal variant to close the coverage gap, deep-ensemble and MC-dropout comparisons, and validation against a second pilot region.

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